

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint Based Problem Solving

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO5 – Naturalization (independent application using OpenCV)P5	Assessment:	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	25% of CIA	Total Marks:	100
CO4 Statement:	Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos. (BTL6)		
Student Name:	Kaviya V	Reg. No.:	192421022
Section / Batch:	Third Year/ B.Tech IT	Date:	25/08/2026

Code of Conduct

I ____KAVIYA V / 192421022____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____Kaviya V____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	

QUESTIONS:

6. Real-Time Object Detection on Mobile Device: A mobile application must:

- A. Detect objects using the camera
- B. Run in real time
- C. Use limited CPU and memory

Design an OpenCV-based pipeline from image acquisition to detection and display.
Justify how your design ensures efficiency under constraints.

ANSWERS:

6. Real-Time Object Detection on Mobile Device

1. Understanding the Problem:

The mobile application must detect objects using the camera while maintaining real-time performance with limited CPU and memory.

The main challenge is to achieve a balance between detection accuracy, processing speed, and resource usage.

2. Proposed OpenCV-Based Pipeline:

Camera → Frame Acquisition → Resize → Preprocessing → Object Detection → Draw Bounding Boxes → Display

Step 1: Image Acquisition

- Capture frames continuously from the mobile camera.
- Process one frame at a time instead of storing multiple frames.

Step 2: Frame Resizing

- Resize the captured frame to a smaller resolution such as 320×240 .
- Smaller images require less memory and processing time.

Step 3: Preprocessing

- Convert the frame into a suitable format for detection.
- Apply only necessary preprocessing to avoid unnecessary computation.

Step 4: Object Detection

- Apply a lightweight OpenCV-compatible object detection model.
- Detect objects and obtain their bounding boxes and confidence values.

Step 5: Display

- Draw bounding boxes and object labels on detected objects.
- Display the processed frame immediately.

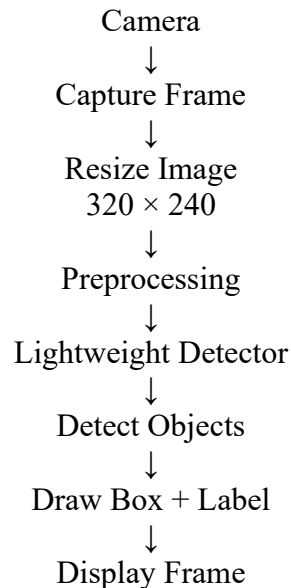
3. Efficiency Under Constraints:

Constraint	Solution
Limited CPU	Use a lightweight detection model and smaller input frames
Limited Memory	Process frames individually instead of storing video frames
Real-time requirement	Avoid unnecessary processing and maintain a suitable frame rate
Processing time	Resize frames before detection

Battery usage	Process only required frames instead of every camera frame when possible
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4. Suitable Design:

A practical pipeline can be:



5. Justification:

- A lightweight detection model with reduced image resolution is suitable because mobile devices have limited CPU, memory, and battery resources.
- Processing smaller frames reduces the number of pixels that the detector must analyze, thereby reducing computational load and latency. Processing frames individually also prevents unnecessary memory consumption.
- However, reducing resolution can slightly reduce detection accuracy, especially for small objects. Therefore, the resolution should be selected as a balance between speed and accuracy.

6. Conclusion:

The proposed Camera → Resize → Preprocess → Lightweight Detection → Display pipeline is suitable for real-time mobile object detection. It achieves efficiency by reducing image size, using a lightweight detector, processing frames individually, and avoiding unnecessary operations.

Final Answer:

A lightweight OpenCV-based pipeline with frame resizing and efficient frame-by-frame processing is the most suitable solution. It reduces CPU and memory usage while maintaining real-time object detection, making it appropriate for resource-constrained mobile devices.